Kansas Fish Ontology

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Abstract

The Kansas Fish Ontology (KFO) is a structured knowledge framework that integrates ecological, taxonomic, and environmental data about fish species across Kansas to support conservation and management. KFO is developed using the Knowledge Acquisition and Representation Methodology (KNARM) and it systematically captures expert knowledge and domain terminology related to species traits, habitats, and environmental dependencies. Through iterative expert consultations and metadata modeling, KFO defines relationships among ecological entities such as species, primary ecosystems, conservation status, and trophic guilds. The ontology was instantiated in GraphDB, enabling graph-based reasoning and visualization. A semi-automated pipeline is developed combining scripting tools and OWL APIs that facilitated ontology construction and ensured alignment between data and expert input. To enhance accessibility, we built the Fish Ontology Visualizer—a web-based interface developed with Vue.js, Flask, and GraphDB that provides interactive exploration of fish-habitat relationships and conservation insights. We evaluated the ontology through SPARQL competency questions that confirmed the ontology's logical consistency, query performance, and utility for ecological decision-making. By transforming fragmented fish ecology data into a machine-interpretable model, KFO offers a scalable tool for researchers and wildlife managers to analyze species distributions, monitor ecological changes, and guide habitat restoration. Future extensions will incorporate dynamic environmental data and predictive analytics, further enhancing its potential as a decision-support system for sustainable aquatic resource management.